

Homework 2 Theory

Q1 The $m \times m$ matrix AA^T is equivalent to the $d \times d$ matrix $A^T A$

$$AA^T = A^T A$$

We compute the eigendecomposition of $d \times d$

$$A^T A = Q D Q^T$$

per the spectral theorem

Where: Q contains eigenvectors
 D contains eigenvalues
and e_i is a basis vector for selecting the i th eigenvector or eigenvalue

We get eigenvalues from the $d \times d$ matrix:

$$(A^T A)(Q e_i) = Q D Q^T Q e_i = Q D e_i$$

We get eigenvalues from the $m \times m$ matrix:

$$(A A^T)(A(Q e_i)) = A(A^T A)(Q e_i) = A(Q(D e_i)) = (A Q)(D e_i)$$

Q2

- a) True - one labeled $x(w)$ must predict y
- b) True - no labels, just predict variance
- c) False - both are matrix operations
- d) During preprocessing you can either remove one of the dependent variables or use PCA to project onto an independent subspace $d \times d$